

Set Up Your Own WEATHER SATELLITE RECEIVING STATION

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We are all familiar with the earth images of clouds and hurricanes taken from the earth-orbiting satellites in our daily weather forecasts. With the modern technology and internet connectivity, now a days, such images are accessible to any individual at the click of a button, or even through apps on mobile phones. However, such images are provided either by government agencies or commercial outlets that offer greater details for the weather forecasters.

Often the images come from geostationary satellites that send the high-speed data at microwave frequencies in the range of 3 to 4GHz in C-band, or even higher frequencies. Thus, the actual antenna and RF electronics at these microwave frequencies is a limiting factor for a common experimenter. Besides special design skills and expertise in the RF and microwave domain, authorisation or licensing is also needed to receive such data or signals.

There are, however, some satellites that provide good-quality weather images of a particular location on earth, which can be freely used for one's own good purpose. The images shown in Fig. 1 and Fig. 2 were received from such satellites. Although these are not very high resolution images, they are sufficient for keeping track of the weather and cyclones for an individual experimenter.

The present technology and software tools that are available as freeware can be used to set up one's own weather satellite receiving station. This article will help you set up one such station for yourself.

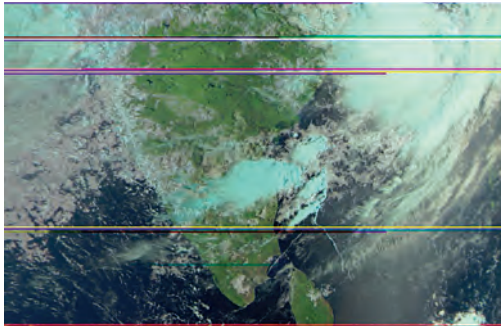


Fig. 1: A high-resolution (1km per pixel) image received from Meteor-M2 satellite by the author

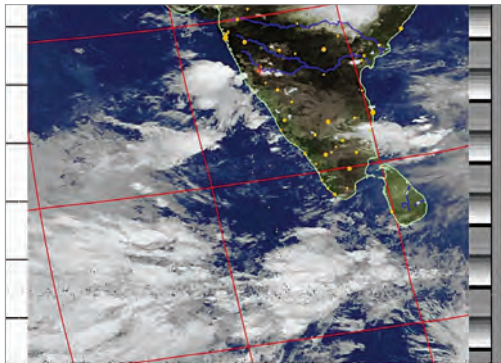


Fig. 2: An image from NOAA15 with some graphics overlay

The basic system

Let us first have a view of the basic system configurations and the type of satellites we can connect to for this purpose. Fig. 3 shows the basic block diagram of a satellite receiving station. Obviously, we need a good antenna, a receiver, and a computer to run the image decoding software. But what's important is to know the satellites that can be used for this purpose. Let's find out!

There are primarily, three types of artificial satellites that orbit the earth. Let us try to understand them.

GEO satellites. Geosynchronous Earth Orbit (GEO) satellites have an orbit of around 36,000km above

the earth's surface. These rotate at the speed of earth and thus appear to be stationary when viewed from earth. The GEO satellites cover one-third of the earth and are often equipped with very high resolution multi-spectral imagers operating in optical to microwave (radio) wavelengths. Since these satellites are stationary relative to earth, it is possible to use high-gain parabolic antennas on earth to receive their microwave signals (refer Fig. 4).

MEO satellites. Medium Earth Orbit (MEO) satellites have an orbit of 10,000 to 15,000km, in between the GEO and LEO satellites in height.

LEO satellites. Low Earth Orbit (LEO) satellites have an orbit of 800 to 1600km and are often non-stationary—or at least they orbit in the sun-

synchronous polar orbits and typically have an orbital period of 90 to 100 minutes. Polar orbit and tracking antenna are shown in Fig. 5.

The National Oceanic and Atmospheric Administration (NOAA) of the USA operates a series of such polar orbiting weather satellites. Of prime focus for us are the NOAA-15, NOAA-18, and NOAA-19 that are currently operational. These satellites transmit low-resolution, typically 4km-per-pixel, 8-bit image data.

The signals are transmitted in what is called Automatic Picture Transmission (APT) protocol in an audible tone with standard frequency modulation (FM) at a carrier in the